

UNIT 4 CATALYSIS, LP 3 4.6 BIOLOGICAL CATALYST

Biological Catalyst - Enzymes

Numerous organic reactions are taking place in the body of animals and plants to maintain the life process. These reactions being slow remarkably catalysed by the organic compounds known as **Enzymes**. All enzymes have been found to be complex protein molecules. Thus: Enzymes are protein molecules which act as catalysts to speed up organic reactions in living cells. The catalysis brought about by enzymes is known as Enzyme Catalysis.

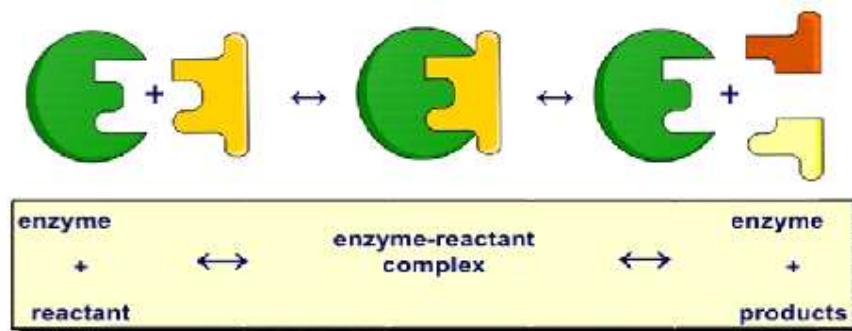
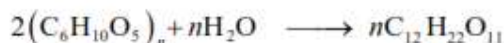
Each enzyme is produced in a particular living cell to catalyse a reaction occurring in that cell. Many enzymes have been identified and obtained in pure crystalline state from the cells to which they belong. However, the first enzyme as prepared by synthesis in the laboratory in 1969.

Enzymes are substances found in biological systems that act as catalyst for specific biochemical process. Enzymes are usually protein or steroid which is synthesized in the living cells of animals and plants. Enzymes catalyze reactions inside organism. Enzymes possess a incredible capacity to carry out complex chemical reactions like hydrolysis, oxidation, reduction etc.

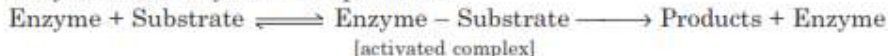
Eg.(i) Amylase is an enzyme which breaks down starch into glucose.



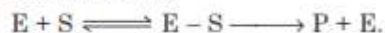
(ii) Diastase converts starch to maltose and maltase converts maltose to glucose



Enzymatic activity can be explained as:



The reaction is abbreviated as:



where E = enzyme, S = substrate (reactant),

ES = activated complex, P = products.

The reaction may be represented pictorially as follows

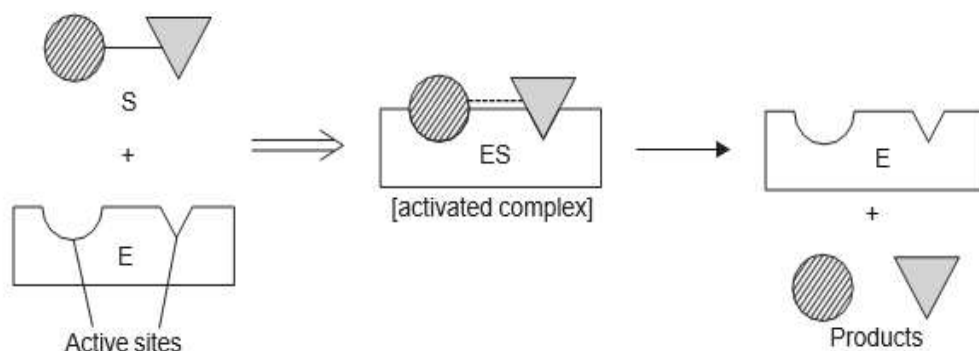


Illustration of the lock-and-key model of enzyme catalysis

Enzymes speed up reactions by lowering activation energy. Many enzymes change shape when substrates bind. This is termed as *"induced fit"*. Enzymes have active sites. The enzyme active site is the location on the enzyme surface where substrates bind, and where the chemical reaction catalysed

Enzymes do not

- Change the equilibrium constant for a reaction
- Change ΔG for a reaction
- Convert a nonspontaneous reaction into a spontaneous reaction.

Factors Affecting Enzyme Activity

a. Enzyme Concentration

- ✓ If we keep the concentration of the substrate constant and increase the concentration of the enzyme, the rate of reaction increases linearly. (That is if the concentration of enzyme is doubled, the rate doubles).
- ✓ This is because in practically all enzyme reactions the molar concentration of the enzyme is almost always lower than that of the substrate.

b. Substrate Concentration

- ✓ If we keep the concentration of the enzyme constant and increase the concentration of the substrate, initially, the rate increases with substrate concentration, but at a certain concentration, the rate levels out and remains constant.



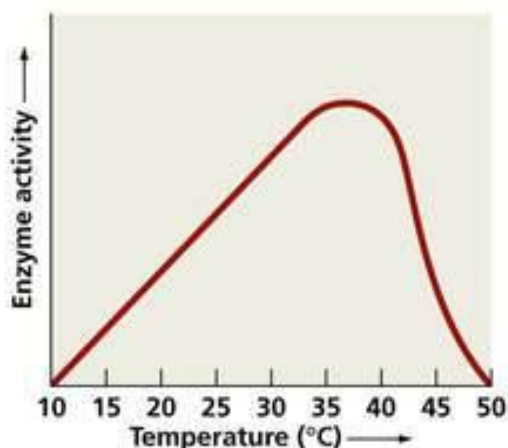
- ✓ So at some point, increasing the substrate concentration does not increase the rate of reaction, *because the excess substrate cannot find any active sites to attach to.*

c. Temperature

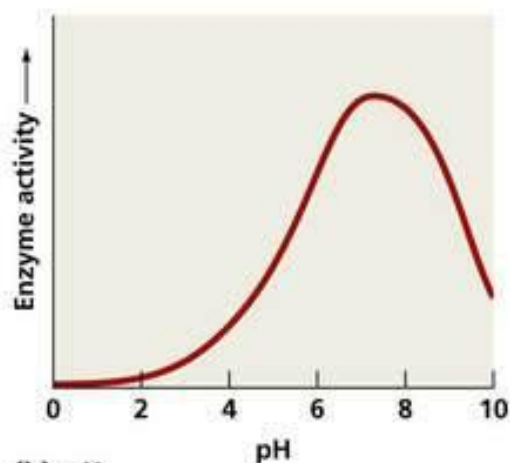
- ✓ For enzyme-catalyzed reactions, like all chemical reactions, rate increases with temperature. However, enzymes are proteins, and at higher temperatures proteins become denatured and inactive. Thus, every enzyme has an optimum temperature.
- ✓ Optimum temperature - the temperature at which enzyme activity is highest-usually about 25°C–40°C.

d. Effect of pH

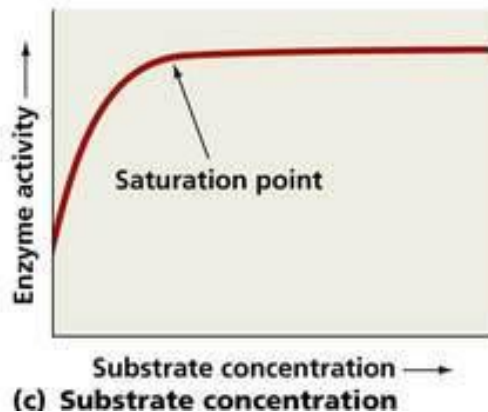
- ✓ Small changes in pH can result in enzyme denaturation and loss of catalytic activity.
- ✓ Because the charge on acidic and basic amino acid residues located at the active site depends on pH. Most enzymes only exhibit maximum activity over a very narrow pH range.
- ✓ Most enzymes have an optimum pH that falls within the physiological range of 7.0-7.5. Notable exceptions are the digestive enzymes pepsin and trypsin. pepsin (active in the stomach) - optimum pH of 1.5. trypsin (active in the small intestine) - optimum pH of 8.0



(a) Temperature



(b) pH



(c) Substrate concentration

Hemoglobin and Myoglobin

Both hemoglobin and myoglobin are globular proteins and share a similar structure that includes a heme group, which contains an iron atom capable of binding oxygen. However, their overall structures differ, especially in their quaternary structure.

1. Hemoglobin:

- **Quaternary structure:** Hemoglobin is a tetramer composed of four polypeptide subunits: two alpha (α) chains and two beta (β) chains. Each subunit contains a heme group.
- **Oxygen binding:** Hemoglobin exhibits cooperative binding, meaning the binding of oxygen to one heme group increases the affinity for oxygen at the remaining heme sites. This allows hemoglobin to efficiently pick up oxygen in the lungs and release it in tissues with low oxygen concentrations.

2. Myoglobin:

- **Tertiary structure:** Myoglobin is a monomer, composed of a single polypeptide chain with a single heme group.
- **Oxygen binding:** Myoglobin has a higher affinity for oxygen than hemoglobin, making it more suited for oxygen storage. It binds oxygen tightly and releases it only when the oxygen concentration in muscles is very low.

Function of Hemoglobin and Myoglobin

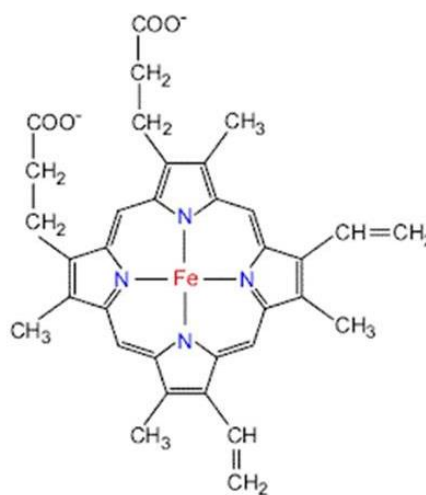
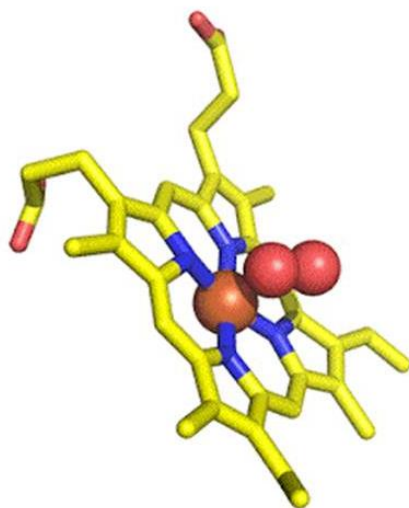
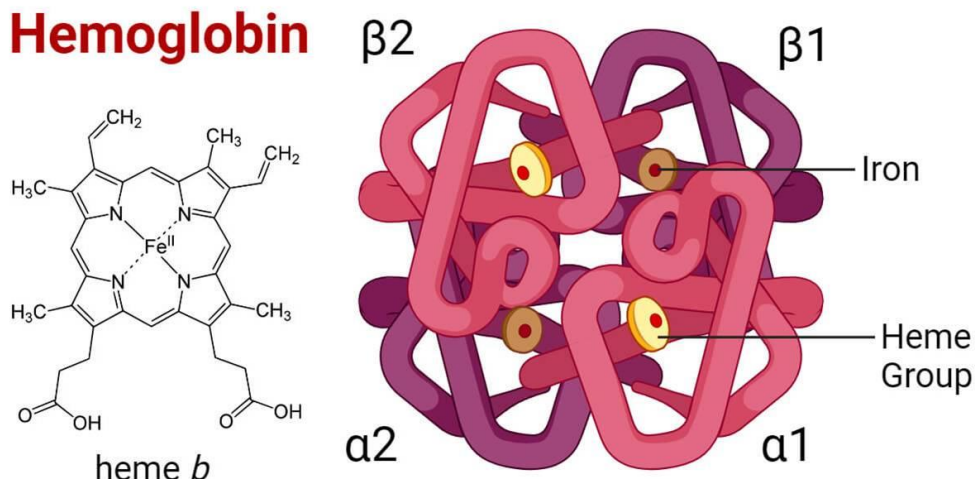
• Hemoglobin:

- The primary function of hemoglobin is to **transport oxygen** from the lungs to various tissues and organs and return carbon dioxide from tissues to the lungs for exhalation.
- Hemoglobin exists in two forms:
 - **R-state (Relaxed):** The oxygen-binding form, where oxygen binds more easily to the heme groups.
 - **T-state (Tense):** The deoxygenated form, where the affinity for oxygen is lower, promoting the release of oxygen into tissues.

• **Myoglobin:**

- The primary function of myoglobin is to **store oxygen** in muscle cells, particularly in **skeletal and cardiac muscles**.
- Myoglobin acts as an oxygen reserve that can be mobilized during periods of high metabolic demand, such as intense exercise, when oxygen consumption in muscles is high.

Hemoglobin



Myoglobin

Differences between Hemoglobin and Myoglobin

Feature	Hemoglobin	Myoglobin
Structure	Tetramer (4 subunits)	Monomer (1 subunit)
Oxygen Affinity	Lower, with cooperative binding	Higher, binds oxygen tightly
Function	Oxygen transport from lungs to tissues	Oxygen storage in muscles
Location	Red blood cells (erythrocytes)	Muscle tissue (skeletal and cardiac muscles)
Oxygen Binding	Cooperative (one oxygen binding increases affinity for others)	Non-cooperative (binds oxygen individually)

Oxygen Binding and Cooperativity

Hemoglobin's cooperative binding means that as one oxygen molecule binds to one of its heme groups, it induces a conformational change that increases the affinity of the remaining sites for oxygen. This property is crucial for hemoglobin's ability to efficiently load oxygen in the lungs (high oxygen pressure) and unload it in tissues (low oxygen pressure).

- In contrast, myoglobin does not exhibit cooperativity. Since it only has one heme group, it binds oxygen in a **hyperbolic curve**, meaning its affinity for oxygen is high and it will hold onto oxygen until the oxygen concentration is very low.

Oxygen Dissociation Curves

- The **oxygen dissociation curve** for hemoglobin is **sigmoidal (S-shaped)**, reflecting its cooperative binding.
- The oxygen dissociation curve for myoglobin is **hyperbolic**, showing a steep increase in oxygen binding as oxygen concentration increases.

Factors Affecting Oxygen Binding

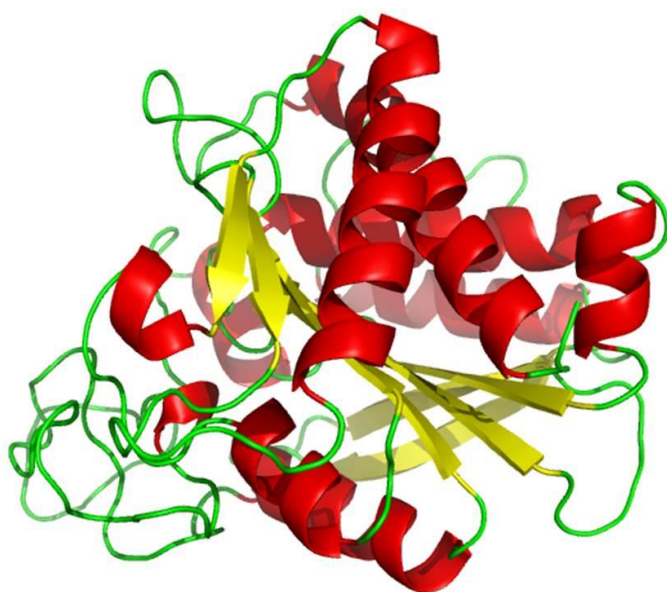
Several factors can influence the binding and release of oxygen by hemoglobin and myoglobin:

1. **pH (Bohr Effect):** A decrease in pH (higher acidity) promotes the release of oxygen from hemoglobin, facilitating oxygen delivery to tissues that are metabolizing and producing acid. Myoglobin is less sensitive to pH changes.
2. **Temperature:** Increased temperature reduces hemoglobin's affinity for oxygen, promoting oxygen release in actively working tissues.
3. **Carbon Dioxide (CO₂):** The presence of carbon dioxide lowers hemoglobin's affinity for oxygen, facilitating oxygen release in tissues that are producing carbon dioxide. This is part of the **Bohr effect**.
4. **2,3-Bisphosphoglycerate (BPG):** BPG binds to deoxygenated hemoglobin and stabilizes the T-state, promoting oxygen release in tissues.

Carboxypeptidase

Carboxypeptidase is an enzymatic protein that plays a crucial role in the digestion of proteins. It belongs to a family of enzymes called peptidases or proteases, which are responsible for breaking down proteins into smaller peptides and amino acids.

Specifically, carboxypeptidase cleaves amino acids from the C-terminal (carboxyl) end of a protein or peptide. This action is important for protein digestion and the regulation of biological processes involving peptides and proteins.



Carboxypeptidase

Types of Carboxypeptidases

There are several forms of carboxypeptidase, categorized based on their substrate specificity and location of action:

1. Carboxypeptidase A (CPA):

- Acts on aromatic or branched-chain amino acids (e.g., phenylalanine, tyrosine, valine, leucine).
- Primarily found in the pancreatic juices, it works in the small intestine to help break down proteins during digestion.

2. Carboxypeptidase B (CPB):

- Acts on basic amino acids (e.g., lysine, arginine).
- Also secreted by the pancreas and plays a role in digestion by targeting peptides containing basic amino acids.

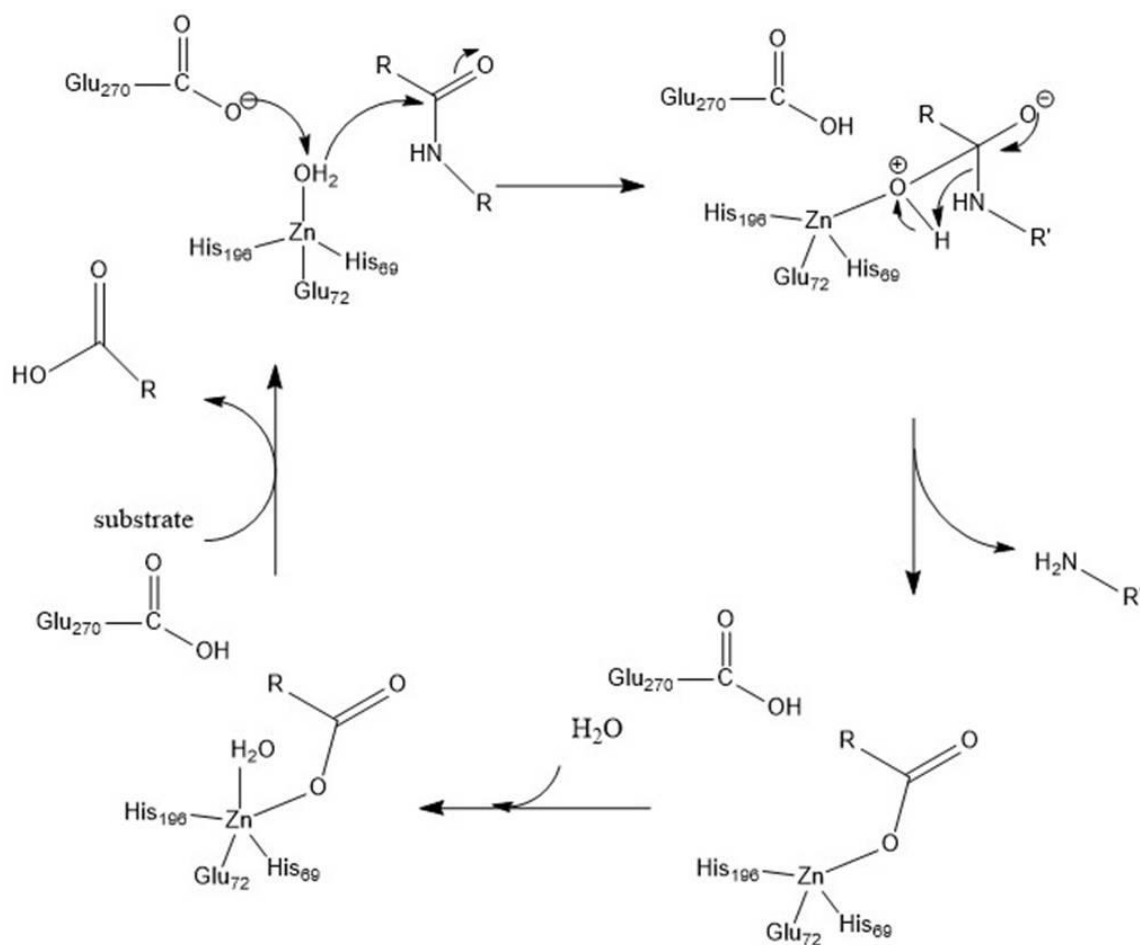
3. Carboxypeptidase C (CPC):

- Found in a variety of tissues, including the **brain** and **kidneys**, it has a more generalized role in breaking down peptides.

Mechanism of Action

Carboxypeptidase functions by breaking the peptide bond between the last amino acid (at the C-terminus) and the rest of the peptide chain. The process involves the following steps:

- 1. Binding:** The substrate (a peptide or protein) binds to the enzyme's active site, positioning the C-terminal peptide bond for hydrolysis.
- 2. Catalysis:** Carboxypeptidase uses a **metal ion** (usually **zinc (Zn^{2+})**) in its active site to assist in catalyzing the hydrolysis reaction. The zinc ion helps stabilize the transition state, allowing the enzyme to break the bond between the C-terminal amino acid and the rest of the peptide.
- 3. Product Release:** The enzyme cleaves the bond, releasing the free amino acid from the peptide chain and leaving a shorter peptide.



Mechanism of carboxypeptidase action

Carboxypeptidase Active Site

- The active site of carboxypeptidase consists of a **zinc ion**, which is essential for its catalytic activity. This metal ion interacts with the peptide substrate and helps in the hydrolysis of the peptide bond.

Role of Carboxypeptidase in Protein Digestion

In the **digestive system**, carboxypeptidase works in conjunction with other digestive enzymes to break down dietary proteins. Here's a typical sequence of events:

- Protein digestion begins in the stomach** with pepsin, which breaks proteins into smaller peptides.



2. **Pancreatic carboxypeptidases (A and B)** are secreted into the small intestine, where they continue breaking down peptides into **free amino acids**.
3. These amino acids can then be absorbed by **enterocytes** (intestinal cells) and transported into the bloodstream for use by the body in **protein synthesis** or energy production.

Key Points

1. Carboxypeptidase is a type of peptidase that removes the last amino acid from the C-terminal end of peptides and proteins.
2. It operates by using a zinc ion in its active site to facilitate hydrolysis of the peptide bond.
3. The two main types, Carboxypeptidase A and Carboxypeptidase B, have different substrate specificities (e.g., aromatic vs. basic amino acids).
4. Carboxypeptidases play a vital role in protein digestion and amino acid absorption.
5. Their activity is regulated by zymogen activation and feedback inhibition mechanisms.
6. Carboxypeptidases are important in biotechnology, clinical research, and protein processing.